

Problem Definition

Predicting a student's (GPA) using a combination of academic and demographic factors of a student.

Objective: To build an end to end regression pipeline that models these relationships. This model helps institutions to proactively identify students who need support based on early semester results.

Dataset Description

The project leverages a student's performance tracking dataset.

Target variable $\rightarrow$ GPA

Features $\rightarrow$ Age, Gender, Study hours, attendance, major, part-time job.

Model Building

Implemented model $\rightarrow$ "Linear Regression"

This algorithm is selected for its high interpretability. In an educational setting it allows administrators to read the exact directional weights (B coefficients)

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_n x_n$$

Model Evaluation

Metrics Tracked: (RMSE) Root Mean Squared Error and R^2 Score (Coefficient of Determination)

The RMSE explicitly identifies the standard deviation of the residuals, showing how close the ~~model~~ model's ~~prediction~~ predictions are to the actual, unscaled GPA.

The R^2 score represents the proportion of variance in ~~student~~ students' GPA that can be explained by input traits like attendance patterns.

Conclusion

The ~~model~~ model pipeline successfully handles structural string mapping, target feature imbalance using scaling constructs. It applies inverse scale transformation.

Limitation $\rightarrow$ Linear models are unable to catch multi-variable ~~and~~ intersections naturally.

Future improvement $\rightarrow$ Introducing tree-based ensemble models (like Random forests)

Reflection:

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While doing the project, got to know ~~about~~ about new functions like `dropna()` and `data.drop_duplicates()`, understood the working of the program, both ~~of~~ logic and mathematical side.

The challenge I faced was I forgot about the inverse transform function, I ~~was~~ was stuck there, the output didn't show any error but logically the GPA was showing wrong output.

Ultimately the project bridged the gap between academic theory and software development. It emphasized that a machine learning workflow is a cohesive, iterative chain of data manipulation, mathematical optimization, and exception ~~handling~~ rather than an isolated set of commands.